Eric Wu

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EDUCATION & HONORS

Cornell University, Duffield College of Engineering **Ithaca, NY**
Candidate for Bachelor of Science in Operations Research and Engineering *Expected May 2029*
Minor in Business in the SC Johnson College of Business
Cumulative GPA: 3.82 / 4.00
Relevant Coursework: Intro Computing, Object-Oriented Programming and Data Structures, Multivariable Calculus, Differential Equations, Linear Algebra, Engineering Operations, Engineering Probability and Stats

EXPERIENCE

Energy and Environment Research Lab - Agrivoltaics Research Program **Ithaca, NY**
Mechanical Intern *Jun 2024 – Aug 2024*
- Developed 3 innovative solar racking prototypes by analyzing 50+ patents and using CAD modeling to enhance durability by 25%
- Optimized solar panel designs through 10+ wind tunnel tests and structural deflection assessments for solar-agriculture integration
- Enhanced solar racking design across 5 installations and integrated an agricultural ecosystem into renewable energy infrastructure

GEC Academy **Shanghai, China**
Summer Program Researcher *Jun 2023 – Aug 2023*
- Applied circuitry design, signal processing, and electrical principles through 5 hands-on projects and 1 proposed team presentation
- Engaged in 20 lectures and discussions with 3 practical applications of circuitry concepts, with emphasis on complex circuit analysis
- Designed and simulated 5+ analog and digital circuits through Multisim and validated performance through theoretical modeling

Oxford AQA **Remote**
Independent Researcher, IEPQ *Dec 2023 – Apr 2024*
- Conducted a comparative analysis of educational robots and software using scholarly databases like IEEE Xplore and JSTOR
- Performed and implemented 5+ qualitative research methods to evaluate the usability, effectiveness, and limitations of various tools
- Analyzed mixed-methods research and integrated feedback along with 1000+ software performance data points to assess solutions

PROJECTS

Dungeon Battle Simulator **Ithaca, NY**
Java *Oct 2026*
- Developed a turn-based dungeon battle game in Java using a multi-level class hierarchy and polymorphism to implement 4 player types, spells, potions, weapons, and randomized turn order
- Managed game state with partitioned arrays and assertion-enforced class invariants and built a validated console interface supporting file-driven input for reproducible testing

Yahtzee **Ithaca, NY**
Python *Apr 2026*
- Engineered a 13-category Yahtzee scoring engine in Python using inheritance and polymorphism, with shared logic in reusable base classes that reduced each new category to under 15 lines of code
- Built an interactive command-line interface (CLI) game supporting dice rolling, strategic rerolls, real-time scorecard updates, robust input validation, and comprehensive error handling, deployable seamlessly across both macOS and Windows operating systems

Automated Glove-Donning Device **Naperville, IL**
Arduino, Fusion 360 (CAD), 3D Printing *Aug 2024 – May 2025*
- Designed a touchless glove-donning device to reduce surgical site infections, using Arduino-controlled linear actuators and UVC disinfection to automate a process that can consume up to 25% of healthcare workers' preparation time
- Analyzed 5 commercial products and 5 patents to define 9 design requirements, narrowed 15+ concepts with decision matrices, and 3D-printed and tested a working prototype across 10 trial runs for accessibility, consistency, and restocking performance

ADDITIONAL INFORMATION

Tools/Skills: Python, Java, Autodesk Fusion 360, Multisim, Soldering, IntelliJ, Eclipse, Pulsar
Languages: Chinese (Fluent), English
Interests: Weightlifting, Albanese Gummies, Snowboarding, Golf, Spontaneous road trips, Cars, Basketball, Apex Legends, Running